1. Title

Failure Prediction AI – Startup & Project Risk Analyzer

Failure Prediction AI
Startup & Project Risk Analyzer

PREDICT RISKS • MITIGATE FAILURES • ENSURE SUCCESS

2. Project Statement

Entrepreneurs, students, startups, and project teams often invest significant time, effort, and resources into ideas without fully understanding the risks that may impact success. Many startup ventures, business plans, hackathon solutions, and college projects fail due to poor market fit, intense competition, budget limitations, operational challenges, or inadequate planning. This project aims to develop Failure Prediction AI, an intelligent decision-support platform that analyses startup ideas, business proposals, college projects, and innovation concepts to predict potential failure risks and recommend mitigation strategies. By leveraging Large Language Models (LLMs), strategic reasoning, market intelligence, risk assessment frameworks, and agentic AI workflows, the system provides data-driven insights into business viability and project success probability. The solution enables users to make informed decisions, reduce uncertainty, and improve the likelihood of successful execution.

3. Outcomes

- AI-powered evaluation of startup ideas, business plans, and project proposals.
- Automated risk scoring and success probability estimation.
- SWOT analysis generation for strategic assessment.
- Competitor and market opportunity analysis (TAM, SAM, SOM estimation).
- Identification of critical business, technical, financial, and operational risks.
- Personalized mitigation strategies and improvement recommendations.
- Interactive dashboard for risk visualization and decision support.
- Comprehensive project viability and feasibility reports (PDF and DOCX formats).

4. Modules to be Implemented

1. Idea & Project Information Collection

- Collect startup, project, business plan, and hackathon idea details.
- Capture target market, budget, competition, resources, and objectives.
- Store proposal information for future analysis in PostgreSQL / Supabase.

2. Market & Competitor Intelligence Engine

- Analyse target market characteristics across multiple industry verticals.
- Identify key competitors and industry challenges.
- Assess market opportunities and growth potential (TAM, SAM, SOM modeling).

3. Risk Assessment & Scoring Engine

- Evaluate business, financial, operational, and technical risks on a 1–5 scale.
- Generate overall risk score and mathematical success probability.
- Prioritize high-impact risk factors.

4. SWOT Analysis & Strategic Evaluation

- Generate Strengths, Weaknesses, Opportunities, and Threats analysis.
- Evaluate project feasibility and sustainability.
- Identify strategic advantages and gaps.

5. Recommendation & Mitigation Engine

- Generate actionable recommendations categorized by priority (Critical, High, Medium).
- Suggest dual mitigation strategies (Preventive and Contingency actions).
- Recommend execution roadmaps and improvement plans via LangGraph agent workflows.

6. Dashboard & Reporting System

- Interactive OLED dashboard for risk visualization.
- Generate detailed feasibility and assessment reports.
- Provide business intelligence and strategic insights with PDF/DOCX exports.

5. Week-wise Module Implementation & High-Level Requirements

Milestone 1 (Weeks 1–2) – Data Collection & Market Intelligence

- Develop startup and project submission forms with input validation.
- Implement market and competitor analysis modules (TAM, SAM, SOM).
- Store project information in PostgreSQL / Supabase.

Milestone 2 (Weeks 3–4) – Risk Assessment & SWOT Analysis

- Develop 5-vector risk scoring engine and success probability formula.
- Generate SWOT analysis matrix automatically.
- Implement project feasibility assessment scoring.

Milestone 3 (Weeks 5–6) – Recommendations & Strategic Reasoning

- Integrate Groq/Llama-3 and OpenAI/Gemini for strategic recommendations.
- Develop mitigation and improvement suggestion engine (Preventive/Contingency).
- Implement LangGraph 6-node autonomous agent workflows.

Milestone 4 (Weeks 7–8) – Dashboard & Deployment

- Build interactive risk analytics dashboard with KPI telemetry.
- Generate comprehensive assessment reports in PDF and DOCX within 5 seconds.
- Perform testing, latency optimization, and cloud deployment.

6. Evaluation Criteria

Milestone 1 (Week 2)

- Successfully collect and process startup/project information.
- Market and competitor analysis generated correctly.

Milestone 2 (Week 4)

- Risk scoring model produces consistent evaluations.
- SWOT analysis generated successfully for submitted projects.

Milestone 3 (Week 6)

- AI recommendations align with identified risks.
- Mitigation strategies generated successfully with clear priority tags.

Milestone 4 (Week 8)

- Dashboard fully operational and responsive.
- End-to-end project risk analysis workflow functional.
- Assessment reports generated within 5 seconds.
- User satisfaction score ≥ 85% during testing and evaluation.

7. Workflow Diagram

Failure Prediction AI – Startup & Project Risk Analyzer Workflow

Phase	Component	Process & Output Description
1. Ingestion	Startup / Idea Submission	Collects Name, Industry, Business Model, Budget, Description, Team, Market.
2. Profiling	Project Profiling Engine	Normalizes attributes; evaluates market size, competitor density, and trends.
3. Market Intel	Market & Competitor Analysis	Calculates TAM ($2.4B), SAM ($850M), SOM ($12M), and historical market trends.
4. Risk Scoring	5-Vector Risk Engine	Quantifies Market, Financial, Competition, Technical, and Operational Risks (1-5).
5. Evaluation	SWOT & Feasibility Engine	Produces Strengths, Weaknesses, Opportunities, Threats, and Feasibility Score.
6. Reasoning	LangGraph Agent + LLM	Executes 6-node reasoning pipeline: Analyzes risks, generates action cards & mitigations.
7. Delivery	Interactive Dashboard & Reports	Renders telemetry KPI cards, area trend chart, and generates instant PDF/DOCX downloads.

8. Architecture Diagram

Multi-Layered Enterprise Architecture

Layer	Technologies / Modules	Core Responsibilities
User Layer	Entrepreneurs, Students, Founders, Analysts	User submissions, interactive queries, risk scenario testing.
Presentation Layer	Streamlit, HTML5, Custom CSS	Cyberpunk OLED dark theme, KPI cards, area charts, filter pills.
Application Layer	Streamlit Controller (app.py)	State management (st.session_state), form validation, routing.
Market Intel Layer	market_analysis.py, Groq API	TAM/SAM/SOM estimation, competitor landscape profiling.
Risk Assessment Layer	risk_engine.py, swot_analysis.py	5-risk vector scoring, SWOT matrix derivation, Feasibility score.
AI Orchestration Layer	LangGraph, StateGraph, LangChain	6-node multi-step sequential reasoning workflow.
LLM Inference Layer	Groq Llama-3 70B / OpenAI / Gemini	Zero-shot JSON reasoning, strategic mitigations synthesis.
Data Persistence Layer	Supabase (PostgreSQL)	Projects, risk assessments, logs, SWOT analysis persistence.
Reporting Layer	ReportLab (PDF), Python-Docx (Word)	On-demand executive report compilation with zero Unicode crashes.

9. Database Schema

Relational Schema: Startup & Project Risk Analyzer

Table Name	Primary Key	Foreign Keys	Key Attributes
users	user_id (UUID)	-	name, email, role, created_at
projects	project_id (BIGSERIAL)	user_id	project_name, industry, business_model, budget, target_market, description
analytics_logs	log_id (BIGSERIAL)	user_id	activity_type, timestamp
competitor_analysis	analysis_id (BIGSERIAL)	project_id	competitor_name, market_position, competitive_risk, analysis_date
swot_analysis	swot_id (BIGSERIAL)	project_id	strengths[], weaknesses[], opportunities[], threats[], generated_at
risk_assessments	risk_id (BIGSERIAL)	project_id	risk_category, risk_score (1-5), risk_description, priority_level
success_predictions	prediction_id (BIGSERIAL)	project_id	success_probability (%), overall_risk_score, prediction_date
recommendations	recommendation_id (BIGSERIAL)	project_id	recommendation_text, risk_mitigation, priority, generated_at
assessment_reports	report_id (BIGSERIAL)	project_id	report_path, generated_at
analytics	usg_id (BIGSERIAL)	project_id	project_type, budget, markets, created_at, generated_at

Document Verification Status: All 9 modules and architectural specifications fully align with Infosys Springboard milestones and criteria. Generated successfully.